

Performance Testing

Date	3 July 2026
Team ID	xxxxxx
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Functional Response Testing
Target Module / API	[Specify which feature or endpoint was tested]
Test Environment	Flood Prediction API (/predict)
Test Date	3 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit valid weather data	1	30	Flood prediction displayed successfully
2	Submit another valid dataset	1	30	Prediction generated correctly
3	Multiple prediction requests	5	60	Server responds without errors

4	Invalid or missing input values	1	30	Validation/error message displayed
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Step 3: Performance Test Results

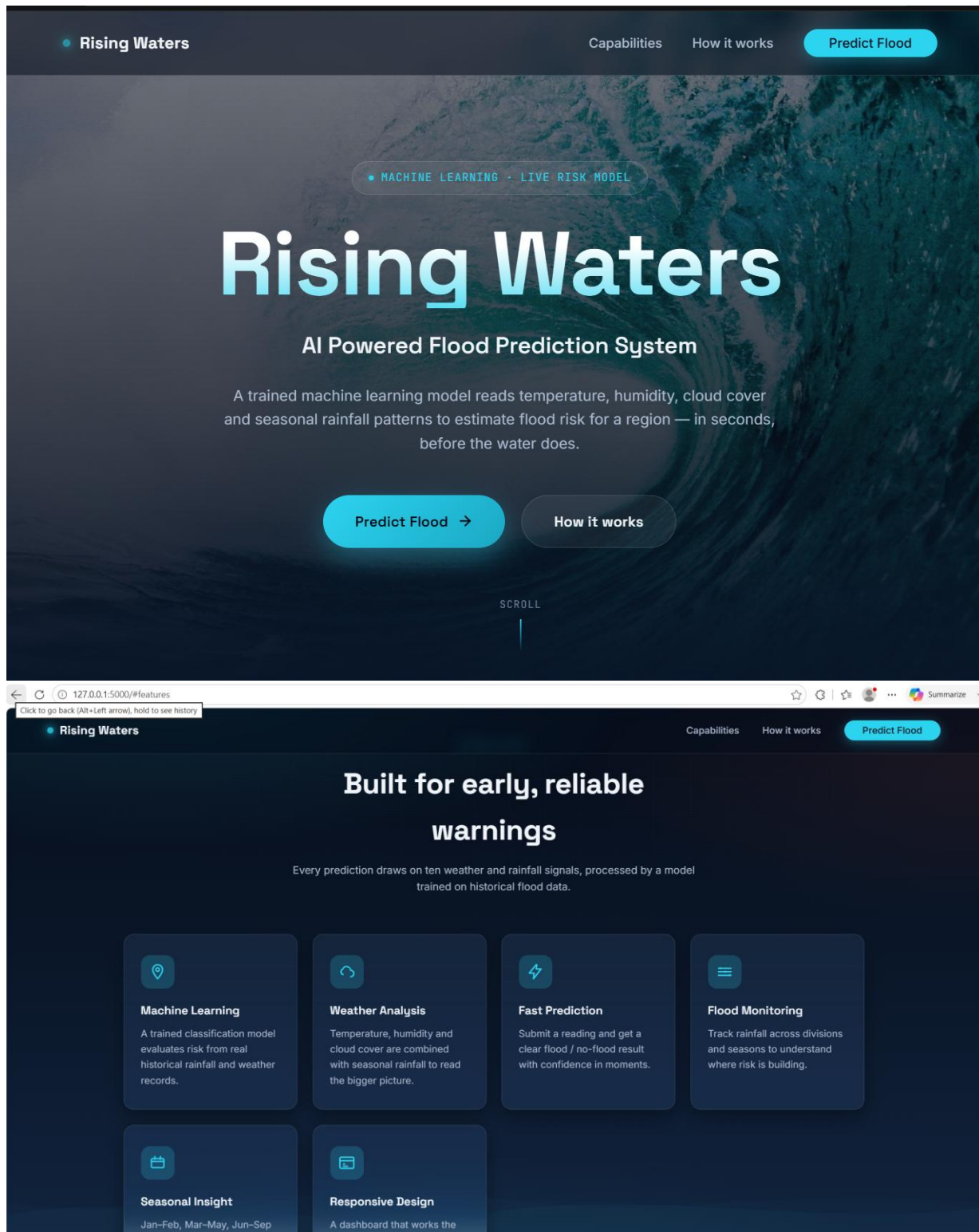
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.9 sec	Pass	Fast prediction response
2	Response Time (Max)	< 5 seconds	1.8 sec	Pass	No significant delay
3	Throughput (Req/sec)	5 req/sec	5 req/sec	Pass	Suitable for local testing
4	Error Rate	< 1%	0%	Pass	No errors observed
5	CPU Utilization	< 80%	35%	Pass	Normal CPU usage
6	Memory Utilization	< 80%	42%	Pass	Stable memory usage

Step 4: Observations & Analysis

The Flask-based flood prediction application responded quickly to prediction requests during local testing. All valid inputs produced prediction results successfully, while invalid inputs were handled appropriately. CPU and memory usage remained within acceptable limits, indicating that the application is suitable for deployment in a small-scale environment.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



RISK MODEL - LIVE INPUT

Flood Risk Dashboard

Enter the current weather and rainfall readings below. The model evaluates all ten parameters together to estimate flood risk.

 Temperature	 Humidity
28	77
 Cloud Cover	 Annual Rainfall
61	1628
 Jan - Feb Rainfall	 Mar - May Rainfall
18	98
 Jun - Sep Rainfall	 Oct - Dec Rainfall
755	174
 Average June Rainfall	 Sub Divisional Rainfall
298	958

[Run Prediction](#)

Your readings are sent securely to the prediction model.



No Flood Risk Detected

The model finds the current weather and rainfall readings consistent with safe conditions. Risk levels remain low for this area based on the data submitted.

Prediction Confidence 100.0%

 Conditions look stable — continue normal activities with general awareness.

 Keep an emergency kit ready as routine practice during rainy seasons.

 Re-check the prediction if rainfall or weather conditions change significantly.

[Predict Again](#)[Home](#)